Img16DatSer.exe
16-bit headerless DataSeries viewer (for Windows, Linux+WinE)

2026-Oct **by Serafim Bochkarev**

The preamble:

The purpose of this application is to gather occasional test of standard GUI-functionalities (especially under WinE: copy/paste, winebrowser, graphic's correctness under Wayland). For mutual benefits:
- a user might have practical functionalities that absent in ~ImageJ (though being trivia and basic);
- the developer might use the reviews in further in further applications (including commercial code).

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1. The introduction

= The executables: in github's repository
"https://github.com/jbstphr/Img16DatSer"

- **Img16DatSer.exe**
  Windows GUI (supports variable modes of StretchBlt).

- **Img16DatSerDX2.exe**
  Windows DX2 (only StretchBlt=halftone);
  <might> work ~visibly faster in older Linux-machines.

= Source code: in github's repository (might need a request to access).
"https://github.com/jbstphr/Img16DatSer_Src"

= The statements:
 - Used sources are:
  -- K&R C RTL, MFC VS 2008, Wikipedia;
  -- exclusively trivial personal-free-time-made source-code.
Beside enlisted above, here is no single explicit (and perhaps implicit) code-snippetfrom another person or organization.
 - This application shares a trivia-subset of functionalities with existing X-Ray SDK.
 - This application excludes all "non-trivia" functionalities (such as "differential view", "directional-blur", "spatial-filters").

1.1 The scope

= Typical files' extensions are:
 - *.bin - some cameras "o naturel";
 - *.raw - ImageJ, AdobePhotoshop (these programs also vary big|little-endianess); some cameras
 - *.img, *.dat - some programs and cameras

= Image format(s):
 16-bit little-endian (signed, unsigned) without header

= Binary's name defines initial state of pixels' interpretation:
 - default : - [signed],
 - *U.exe, *u.exe: - [unsigned].

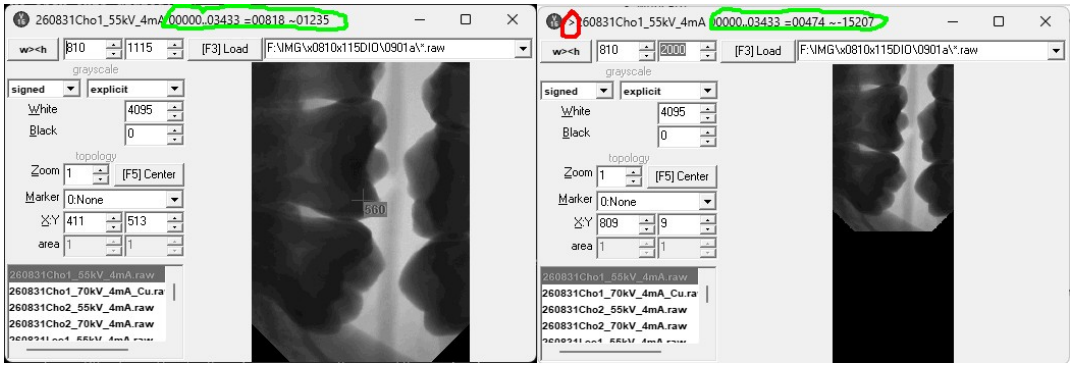
= Specific features for "comprehend and analysis" of data-series
 - Profile-gutter auto-synchronized with image-view (by spatial location and grayscale); This UI gives ordinary person a chance to evaluate "image's information in ~Shannon's definition" such logically as professional radiologist (whose eyes distinguish ~>100 gray_levels) does.
 - Natively debugged with WinE (executes Nautilus-file-manager, copies via WinE/Linux clipboard).

= The main functionality of this utility:
 - read 16bit little-endian pixels from image_file,
 - hold the pixels in memory as 32bit signed ints,
 - displays those as 8bit grayscale.

the input:
= section [View] in ini-profile
 - this_exe_directory/%IniProfileOfSpecificSDK% (if one exists),
 - this_exe_directory/this_exe_name.ini (otherwise);

= optional cmdline
 - "w h wildcard",
 - "wildcard",
 where wildcard composes as "directory/fname_wildcard" (e.g. ~"256 256 *.dat", ~"640 480 z:/img/1????.bin", ~"/home/user/img/*.raw").

window-title:
- <prefix> file_name min..max =median ~average
- '?'="no image", >="[w]*[h]>file_size",...



1.2 The development tools

= The projects are :
- EProg2008.vcproj : VC- build-toolchain (MS VS 2008)
- EProg.vcxproj : MS- build-toolchains (tested MS VS 2017,2002,2026)

= The character sets :
Assuming the WinE, build this project ~always as UNICODE. MBCS is possible for Windows (but ~has no purpose).

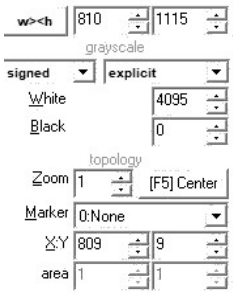
1.3 Usage of VSCode

= File .vscode/c_cpp_properties.json

```
// this is rather a stub to have minimalistic code-highlights in VSCode
// and one-click build (when started under VS-developer-comman-prompt)
{  "version": 4,
   "configurations": [
      { "name": "Win64",
        "includePath": ["${workspaceFolder}/."],
        "defines": ["_WIN32"],
        "compilerPath": "cl.exe",
        "cStandard": "c17",
        "cppStandard": "c++17",
        "intelliSenseMode": "windows-msvc-x64",
        "configurationProvider": "ms-vscode.cmake-tools",
        "browse": {
           "path": ["${workspaceFolder}/**"],
           "limitSymbolsToIncludedHeaders": true
        }
      }
   ]
}
```

= There is no known tactic to make a robust CMakeLists.txt for a typical MFC-project.

2. The main View



2.1 View-controls

```
= [w><h]          :   - swaps the width,height

= [Load]          :   - browse & load image_files

= [combo box] (interprets pixels in image_files)
  - signed        :   - default -32768..32767
  - unsigned      :   - 0..65535 (when named as "View16U.exe")

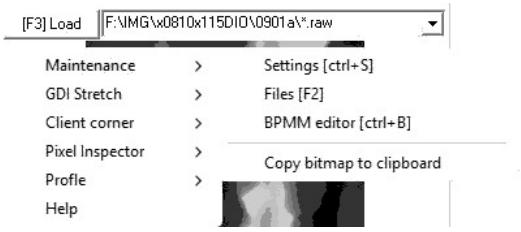
= [combo box] (gray_gamut)
  - explicit      :   - black=>RGB(0,0,0), white=>RGB(255,255,255)
  - THist         :   - cut 0.1% of pixels
  - inv THist     :   - cut 0.1% of pixels and invert

= [Zoom]          :   - magnifies frame's rectangle versus displayed one

= [combo Marker]:   - draws an accessory alongside image

= [edits X,Y]     :   - Marker's location

= [area]          :   - Marker's size
```



2.2 Pop-up menu

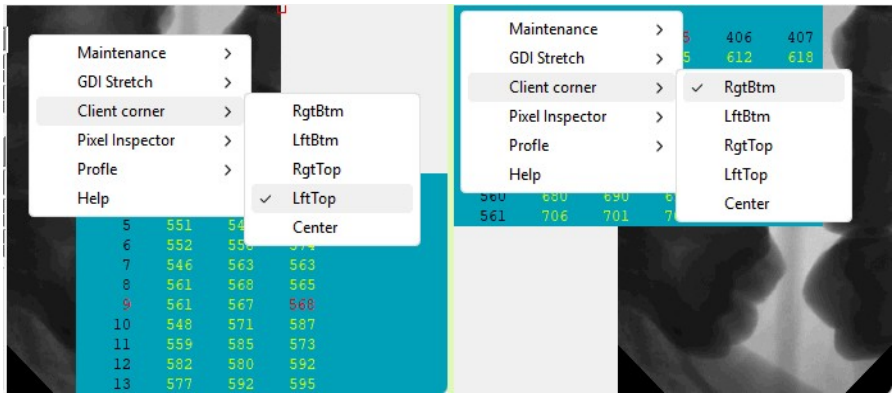
Appears by [F9], right mouse btn

```
      Maintenance
= the dialogs (show particular help by [F1])
  - Settings    :    - UI-settings,
  - Files       :    - operate with items of files' list,
  - BPMM editor:     - operate with BadPixels'Mask-overlays.

= the commands
  - Copy bitmap to clipboard,
  - Copy ROI text to clipboard,
  - Copy recent stats to clipboard.
  - [Execute]; the title might be 'Windows Explorer' or 'WinE Browser'.

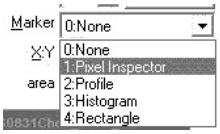

      GDI-stretch selection <when allowed>
- Fast                :    - StretchBlt(...COLORONCOLOR,)
- Halftone            :    - ...HALFTONE (the only option in DirectX-build),
- White|Black details:     - ...WHITEONBLACK|BLACKONWHITE.

      Client corner's selection
- to locate the image (when aspect_ratio reduces client_area),
- to locate accessories in ~opposite corner.
```



Pixel Inspector's selection
- 5x5 | 7x7 | etc.

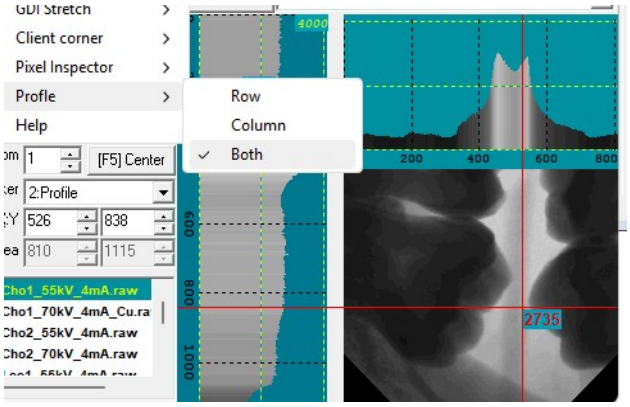
Profile's selection
- Row | Column | Both.



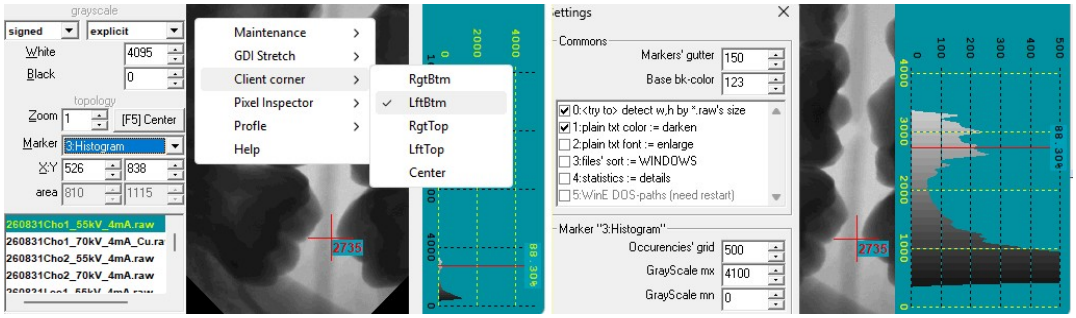
2.3 The markers

1:Pixel inspector
- ROI-center is view's spin-edits [X,Y]
- ROI-area is selection in pop-up menu

2:Profile
- the lines Row|Column|Both follow selection in the pop-up menu
- the grids follow the view

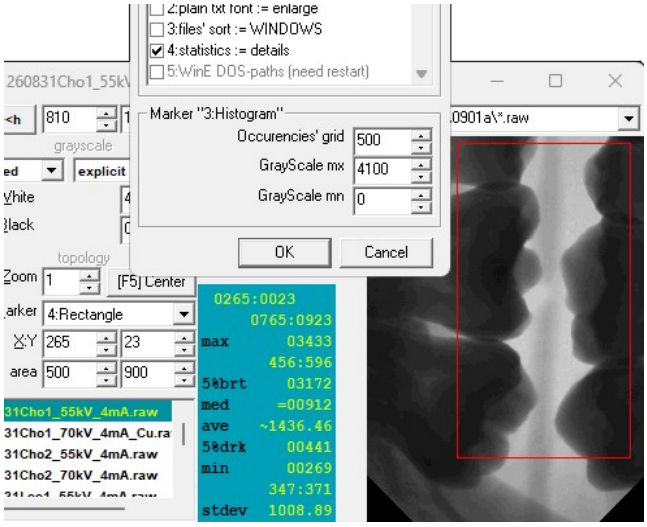


3:Histogram
- the grids follow [Settings]

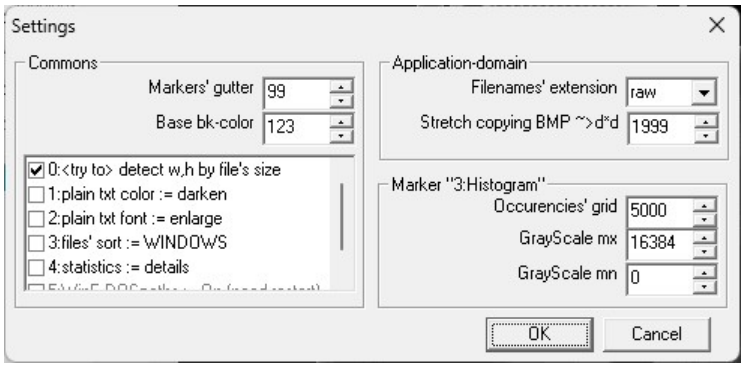


4:Rectangle

- ROI-lft_top is view's [XY]
- ROI-area is view's [area
- presenting statistics follow [Settings]



3. Modal dialog [Settings]



3.1 settings "Commons"

= Markers' gutter ~50..200
<in pixels> width of markers 2:Profile, 3:Histogram.

= Base bk-color ~50..200
<in LSB> brightness of drawing accessories.

```
= Bit_flags
  - 0:detect w,h by file's size" - affects [Load];
the order of checks:
 a) the same FileSize in stored wildcards,
 b) if FS mod sqrt(FS)=0 then w:=h:=sqrt(FS),
 c) if FS mod w=0 then h:=FS div w,
 d) if FS mod h=0 then w:=FS div h.
  -  :"plain txt ... := ..."      - affect drawing accessories,
  - 3:"files' sort := WINDOWS"    - default is alphabet_order,
  - 4:"statiscics := details"     - affects markers 3:Histogram, 4:Rectangle,
  - 5:"WinE DOS-paths := On"      - show DOS-paths, force WinExplorer's stub
(affects under WinE).
```

3.2 settings "Application-domain"

```
= Filenames' extension:
  - *.raw ~ ImageJ, AdobePhotoShop, medical camers
  - *.bin ~ industrial cameras
  - etc

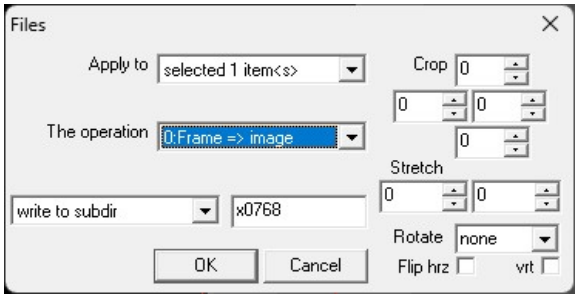
= Stretch copying BMP ~>d*d
  stretch CF_DIB for clipboard if exeeds this area (this mainly for WinE's
limitation). WinE might don't copy CF_DIB ~>4MB (i.e. set "d~<1999" for
grayscale 8-bit BMP in this application).
In Windows:
  - this adhoc limitation might be spractical;
  - one might assign large value (d~>9999) and copy large BMPs via clipboard
exactly.
```

3.3 settings "marker 3:Histogram"

```
= Visible axes:
  - abscissa  - by "GrayScale_mn..mx",
  - ordinate  - by "0..Occurencies_grid".
```

4. Modal dialog [Files]

```
Terminology:
  - 'Frame' - inp image 'as is'
  - 'Image' - transformed 'Frame'.
```




```
      Apply to:
= current selection in files'list,
= whole files'list.

      The operations:
= 0:Frame => Image (in the following order):
  - Crop margins
  - Stretch to specified WxH
  - Rotate
```

```
  - Flip
  - Write ./SubDir/image_file or overwrite existing image_file

= 1:Write simple average frame '_ave'.

= 2:Frame => BMP (cut grayscale image_file<s> to 8-bit-gray *.BMP).

= 3:Delete file<s>.

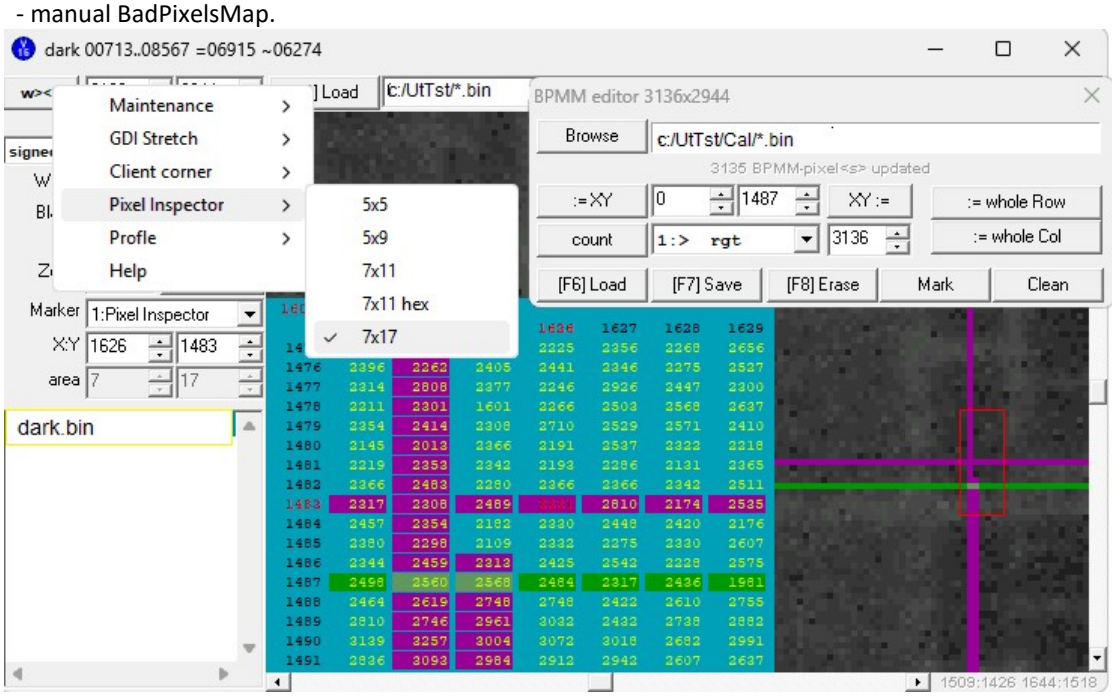
= 4:OS-specific 'open'
  - Invokes ~ShellExecute('open',image_file).

= 5,6:Hide <un>selected file_name<s> in the files'list
  This is to compare histograms, profiles, etc of two image_files, when
those aren't neighbors in files'list initially.
  - a) <by this operation> make them neighbors in the list,
  - b) <by arrow_keys up|down> move files'list_caret and watch quantitative
changes between two charts or between two image_files.") ); }
```

5. Dialog [BPMM-editor]

Infrared cameras, flat-panel X-ray -detectors, industrial optical cameras shall equalize individual $P_{ixelhoton}T_{ransfer}C_{urve}s$. The PTC of aliased pixel can't be properly equalized (by it's currrent value and pre-captured calibration-data alone). Then display of aliased pixel need a 2D-filtration (involving good neighbors). Pixel's alias could be:
= transient, where the fact of alias is detected (and then fixed) run-time;
= permanent, where the fact of alias is:
- detected automatically (from calibration data-series) in advance;
- marked manually in advance.

Respectivelly, calibration-data of those cameras <might> incapsulate two overlay-files:
- automatic BadPixelsMap,
- manual BadPixelsMap.



This dialog shows overlay **BadPixels' map** <automatic>. Shows & edits overlay **BadPixels' map manual**.

At open this dialog:
= <by current loaded image> assigns overlay's dimension,
= <tries to> loads the overlays from SDK-specific location.
= displays status line :
 ~ e.g. "BMP NNNNN pixels, BPMM doesn't exist",
 ~ e.g. "BMP doesn't exist, BPMM MMMMN pixels".

```
Editing area (marked by green color in UI-topology):
- BPMM_org    two spin_edits (x & y) specify 2D-point;
- BPMM_cnt    combo_box & spin_edit specify direction & count.

The actions:
[Browse]             selects location of loading overlays;
[:= XY]              BPMM_org := Marker_XY;
[XY :=]              Marker_XY := BPMM_org;
[count :=]           assigns BPMM_cnt by Marker_XY;
[:= whole Row]       assigns BPMM_org, BPMM_cnt by Marker_XY;
[:= whole Col]       -"-;
[Load]               loads the overlays (abandons changes);
[Save]               saves BPMM in file;
[Erase]              fills BPMM by 0 ([Erase]+[Save] delete BPMM.raw);
[Mark]               fills editing area by 1;
[Clean]              fills editing area by 0.
```

Pixels' colors and text-background:

```
- dark magenta  color -    <>0 in BPM,
- light magenta color -    <>0 in BPMM,
- green         color -    within "editing area".
```

To enhance visibility of ~pixel-size-details on specific monitor:
 - utilize GDI-stretch selection ~= "*White|Black details*" in the pop-up menu (perhaps prefer GDI-build against GPU-build);
 - adjust "base bk-color" in [Settings].

6. An example of visual comparison of two files in the same directory.

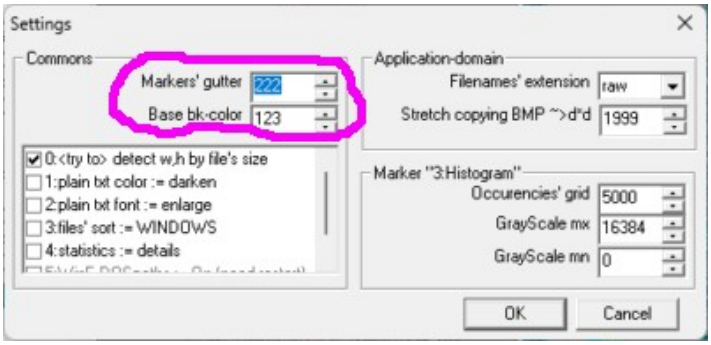
This is an example to comprehend visually **morphological & statistical differences between two <neighbor> images of the same dimension in the same directory**.

Assume a data-series that contains either images and transient image-processing-frames (~derivations, laplacians). A debug-dumper traced:
 - "0Inp" - input image
 - "6Ca_Ave" - a "scrape-laplacian" of ~small magnitude
 - "6Ca_Lap" - a "scrape-laplacian" of ~large magnitude

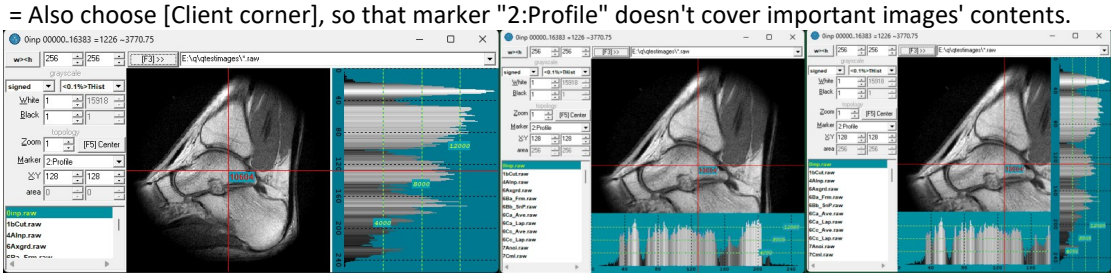
The "0Inp" is <typically> unsigned (used ~<=15 bits) and then loads the same as [unsigned] or [signed]. The "scrape-laplacians" are signed.

6.1 Initial GUI-settings

= Adjust [Markers' gutter], so that marker "2:Profile" has convenient sze:
 - [F9] or right_mouse_button;
 - in pop-up menu click `[Maintenance]/[Settings]`.

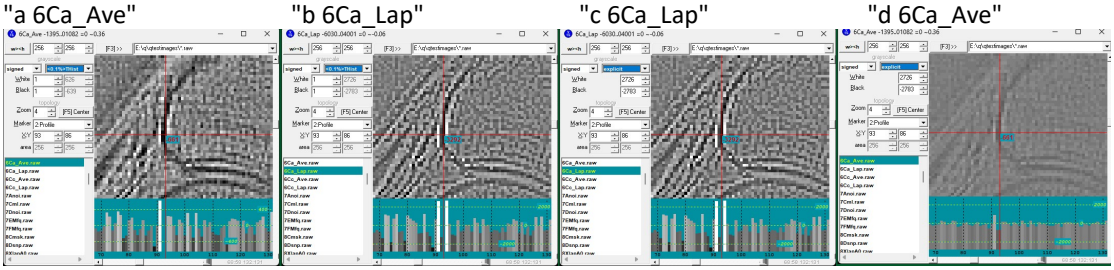


= Also choose [Client corner], so that marker "2:Profile" doesn't cover important images' contents.



6.2 Visually comprehend two laplacians of different magnitude

"a 6Ca_Ave" "b 6Ca_Lap" "c 6Ca_Lap" "d 6Ca_Ave"



= compare morhology:
- select grayscale "**<0.1%> THist**" (truncate histogram in 0.1%);
- comprehend image-presentations "a", "b".

= compare magnitude:
- select laplacian of bigger magnitude ("c");
- select grayscale "**Explicit**" ([White],[Black] hide percentage-spin-edits and enable spin-edits of explicit values; the image-presentation itself remains unchanged);
- view laplacian of smaller magnitude ("d") at the same explict [White], [Black].

7. An example of visual comparison of ~"the same file" in consequent data-series

This is an example to comprehend visually **morphological & statistical differences between ~"the same" images from diffrenet directories**.

Assume e.g., that F:/qq/q0/*.raw & F:/qq/q1/*.raw are two variants of (a similar) image-presentation-enhancement.

An obvious option is to copy specific image<s> "*F:/qq/q1/*.raw*" as ~"*F:/qq/q0/*1.raw*" and then visually compare neighbor items "*F:/qq/q0/*.raw*". However such copying isn't convenient frequently.

7.1 Option A (two view-windows)

= Execute the program;
- load one of the *.raw,
- set suitable grayscale,
- set suitable window size ,
- close program's dialog.

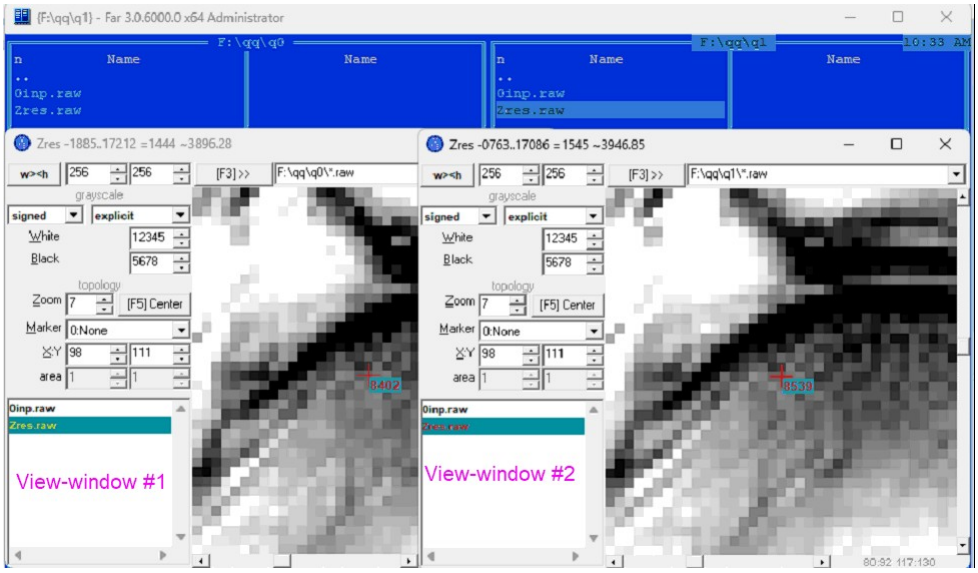
Here the program remembers windows' position & settings in initialization profile

= Execute the program again;

Here the program loads the recent *.raw
- set suitable [Zoom], [Marker XY],
- click "[F5] Center".

= Execute the program again;
- load another *.raw,
- set identical [Zoom], [Marker XY],
- click "[F5] Center".

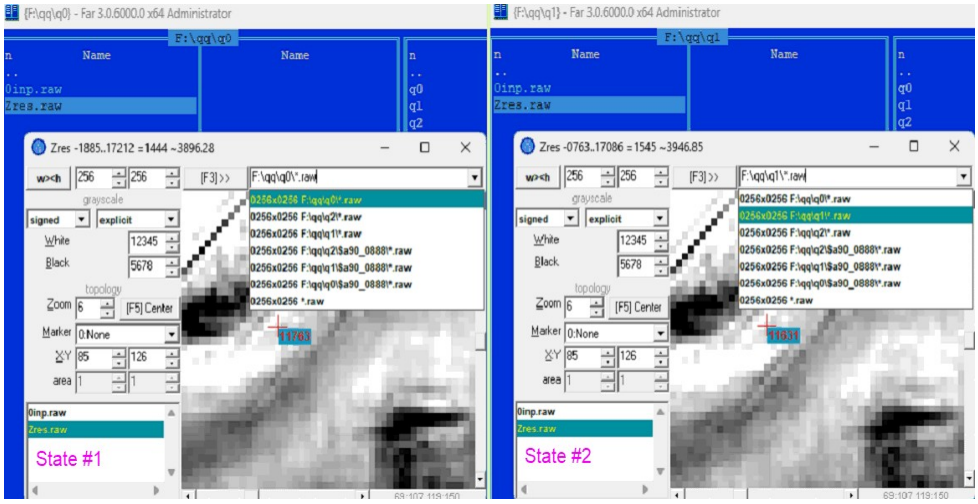
= Compare the views visually.



7.2 Option B (the same view-window)

= Execute the program;
- Load interesting F:/qq/q0/*.raw,
- Load the counterpart F:/qq/q1/*.raw.

= Drop-down the wildcard's combo-box;
- By arrow-keys [up], [down] switch between the directories;
- Watch how the views change against each other.


State #1
State #2